

Nerve classifications

Dr Balapala

*Introduction:

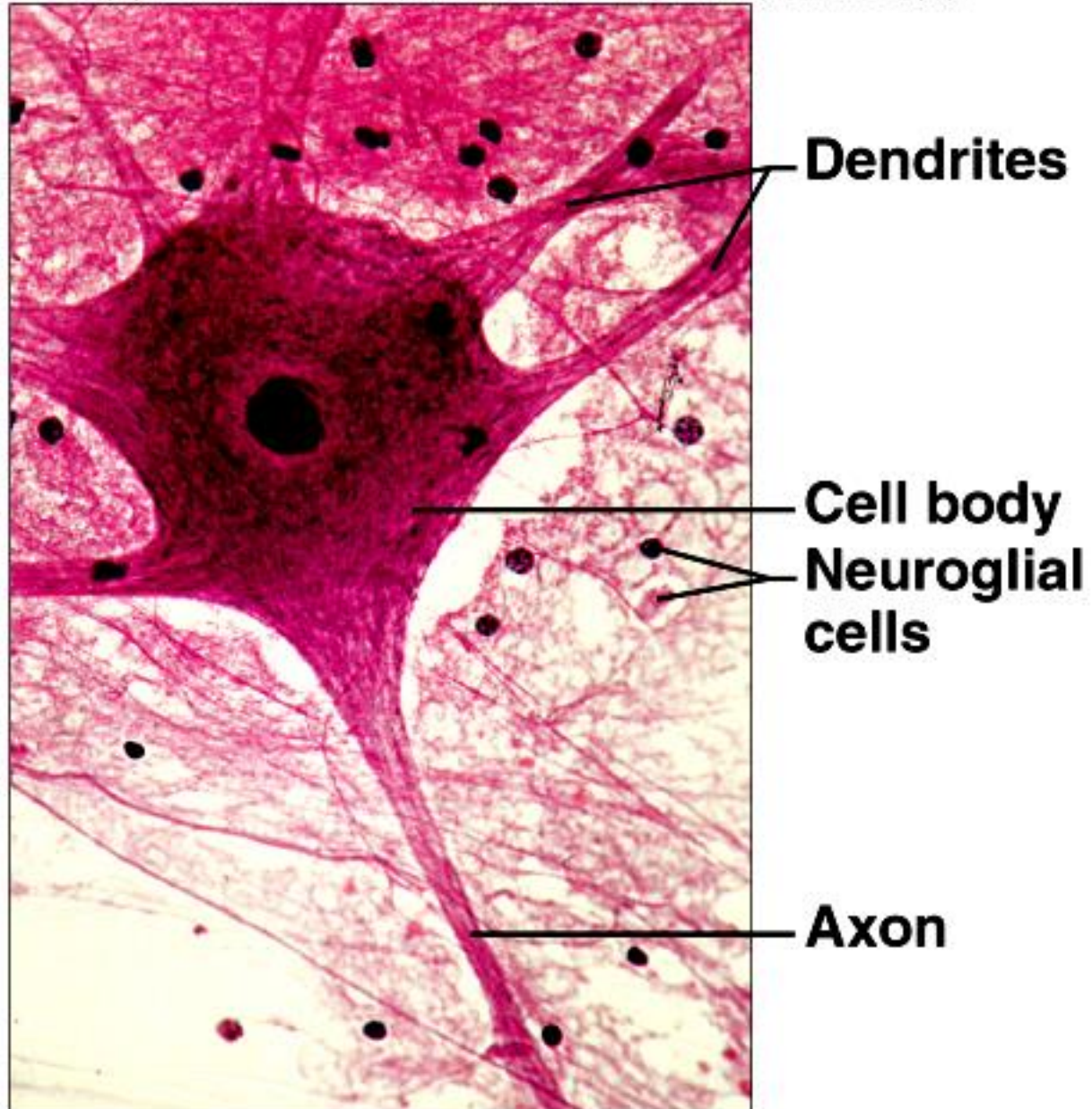
- A. The nervous system is composed of **neurons** and **neuroglia**.
 - 1. Neurons transmit nerve impulses along nerve fibers to other neurons.
 - 2. Nerves are made up of bundles of nerve fibers.
 - 3. Neuroglia carry out a variety of functions to aid and protect components of the nervous system.

- B. Organs of the nervous system can be divided into the **central nervous system** (CNS), made up of the brain and spinal cord, and the **peripheral nervous system** (PNS), made up of peripheral nerves that connect the CNS to the rest of the body.

- C. The nervous system provides sensory, integrative, and motor functions to the body.
 - 1. Motor functions can be divided into the consciously controlled somatic nervous system and the unconscious autonomic system.

✧ General Functions of the Nervous System

- A. Sensory receptors at the ends of peripheral nerves gather information and convert it into nerve impulses.
- B. When sensory impulses are integrated in the brain as perceptions, this is the integrative function of the nervous system.
- C. Conscious or subconscious decisions follow, leading to motor functions via effectors.



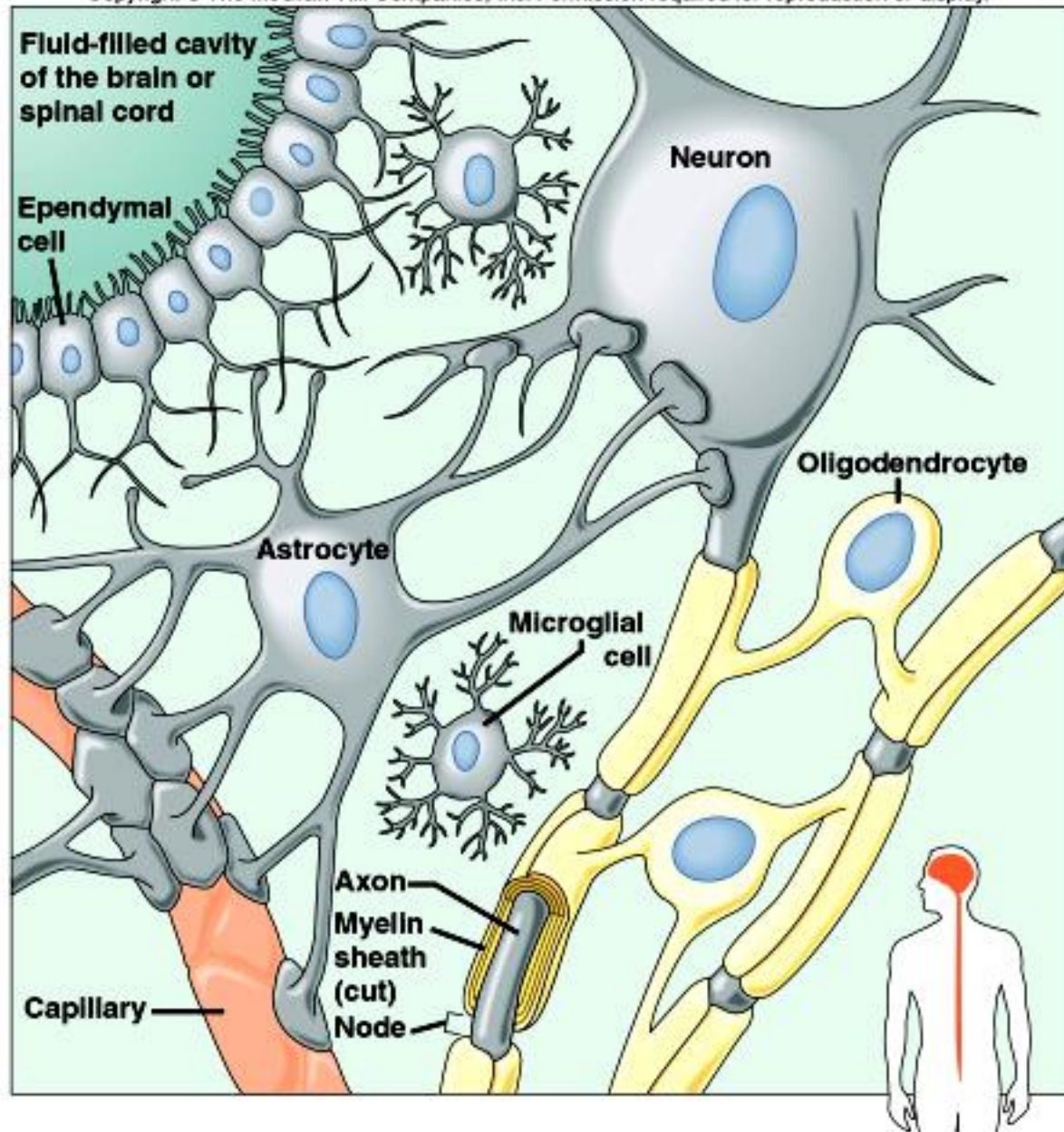
★ Neuroglial Cells

A. Classification of **Neuroglial Cells**

1. Neuroglial cells fill spaces, support neurons, provide structural frameworks, produce myelin, and carry on phagocytosis. Four are in the CNS and the last in the PNS.
2. Microglial cells are small cells that phagocytize bacterial cells and cellular debris.

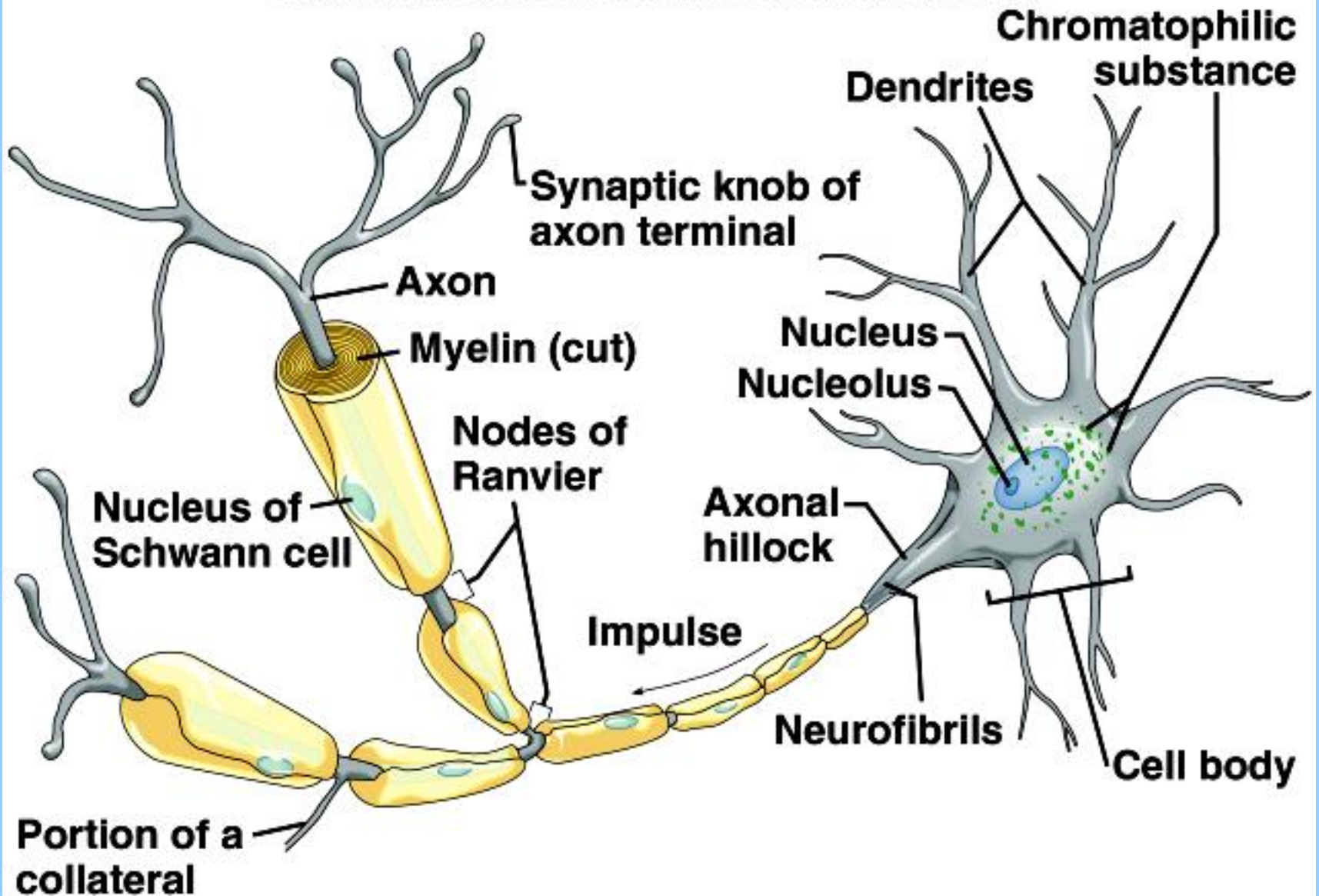
3. Oligodendrocytes form myelin in the brain and spinal cord.
4. Astrocytes are near blood vessels and support structures, aid in metabolism, and respond to brain injury by filling in spaces.

5. Ependyma cover the inside of ventricles and form choroid plexuses within the ventricles.
6. Schwann cells are the myelin-producing neuroglia of the peripheral nervous system.



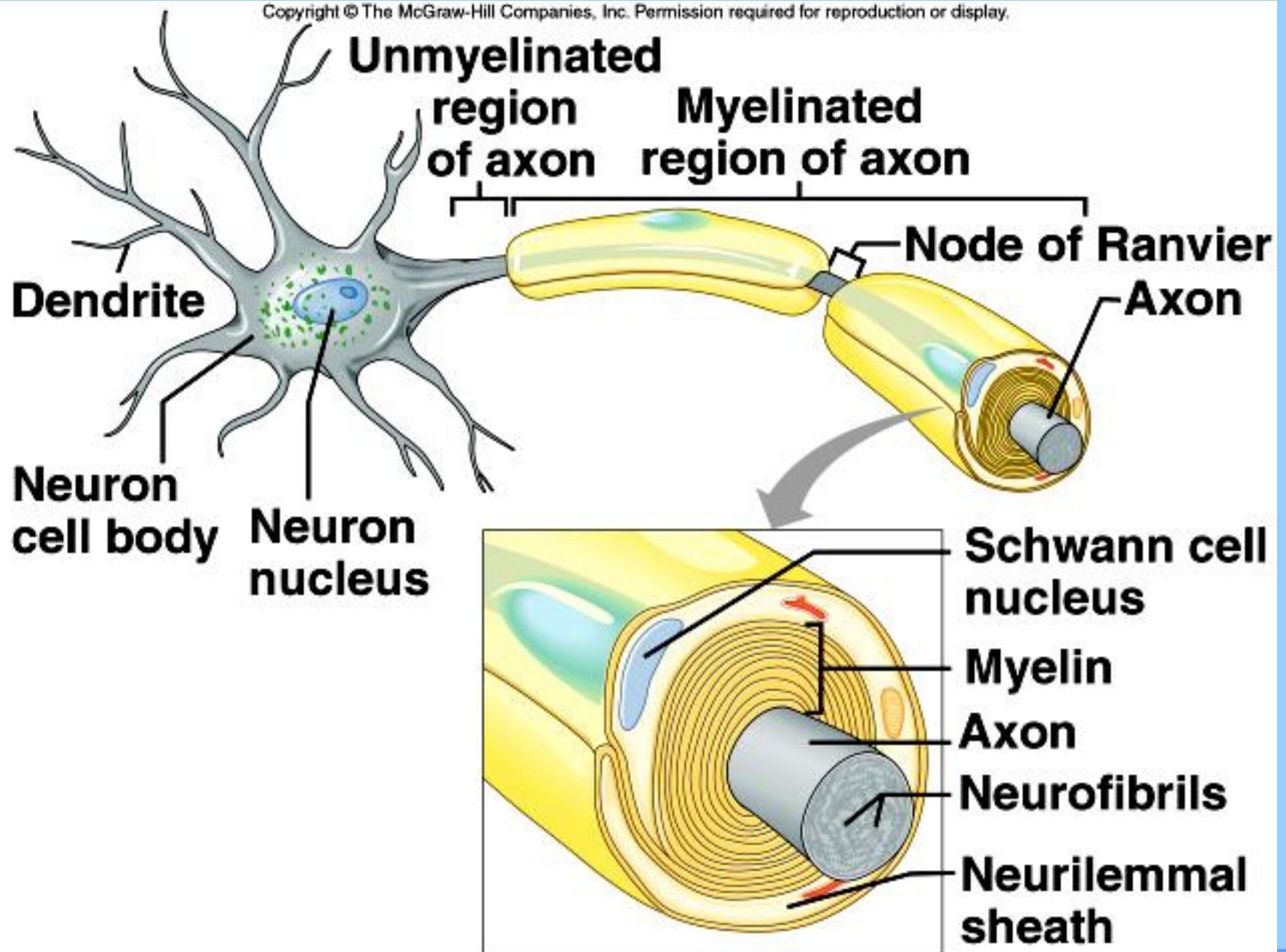
✴ **Neuron Structure**

- A. A neuron has a cell body with mitochondria, lysosomes, a Golgi apparatus, chromatophilic substance (Nissl bodies) containing rough endoplasmic reticulum, and neurofibrils.



- B. Nerve fibers include a solitary axon and numerous dendrites.
 - 1. Branching dendrites carry impulses from other neurons (or from receptors) toward the cell body.
 - 2. The axon transmits the impulse away from the axonal hillock of the cell body and may give off side branches.

3. Larger axons are enclosed by sheaths of myelin provided by Schwann cells and are myelinated fibers.
 - a. The outer layer of myelin is surrounded by a neurilemma (neurilemmal sheath) made up of the cytoplasm and nuclei of the Schwann cell.
 - b. Narrow gaps in the myelin sheath between Schwann cells are called nodes of Ranvier.

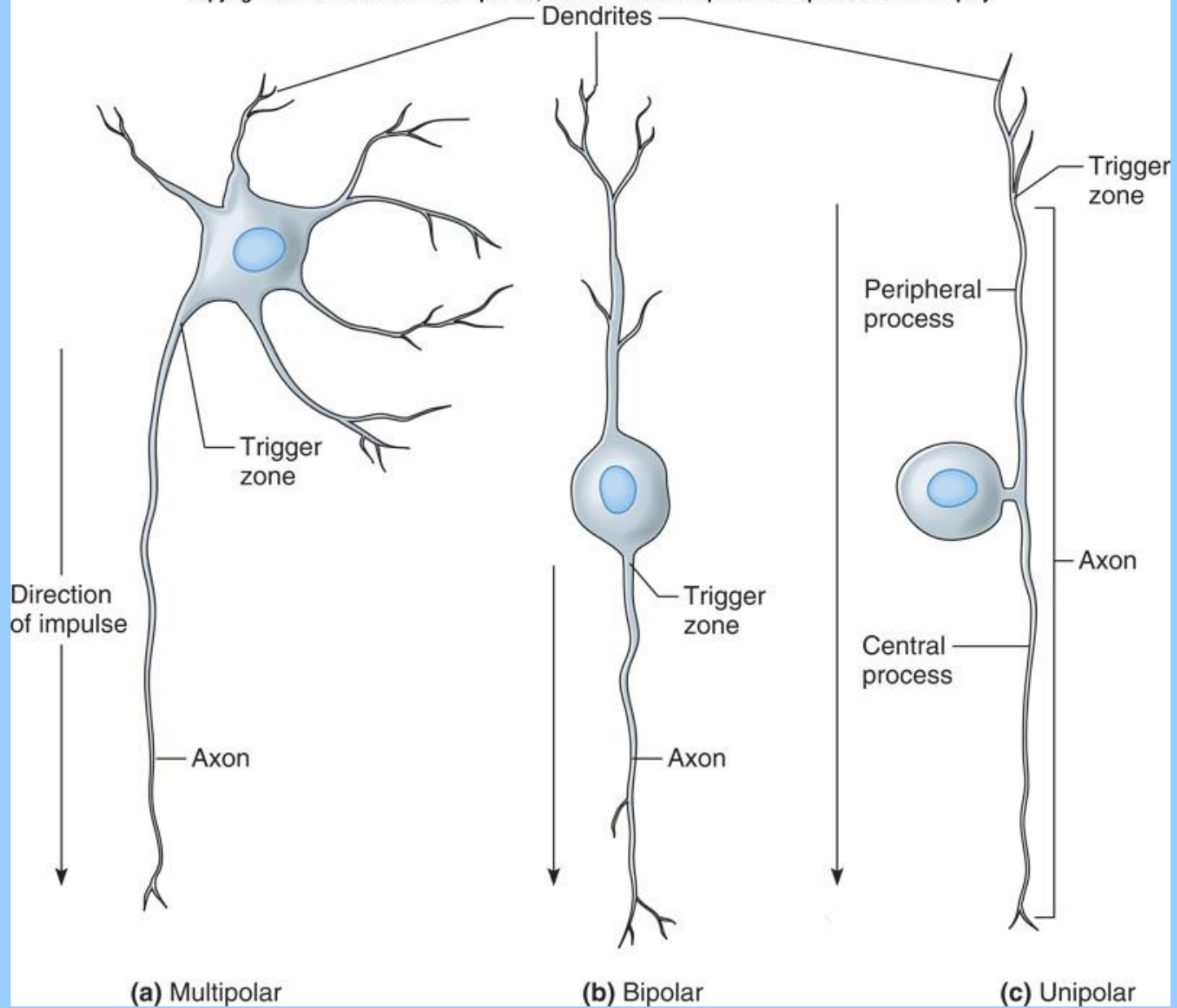


4. The smallest axons lack a myelin sheath and are unmyelinated fibers.
5. White matter in the CNS is due to myelin sheaths in this area.
6. Unmyelinated nerve tissue in the CNS appears gray.

7. Peripheral neurons are able to regenerate because of the neurilemma, but the CNS axons are myelinated by oligodendrocytes, thus lacking neurilemma, and usually do not regenerate.

Classification of Neurons

- A. Neurons can be grouped in two ways: on the basis of structural differences (bipolar, unipolar, and multipolar neurons), and by functional differences (sensory neurons, interneurons, and motor neurons).



B. Classification of Neurons

1. **Bipolar neurons** are found in the eyes, nose, and ears, and have a single axon and a single dendrite extending from opposite sides of the cell body.
2. **Unipolar neurons** are found in ganglia outside the CNS and have an axon and a dendrite arising from a single short fiber extending from the cell body.

3. **Multipolar neurons** have many nerve fibers arising from their cell bodies and are commonly found in the brain and spinal cord.
4. **Sensory neurons (afferent neurons)** conduct impulses from peripheral receptors to the CNS and are usually unipolar, although some are bipolar neurons.

5. **Interneurons** are multipolar neurons lying within the CNS that form links between other neurons.
6. **Motor neurons** are multipolar neurons that conduct impulses from the CNS to effectors.

Classification

- There are three broad functional classifications of nerves – **sensory (afferent), intermediate and motor (efferent)**.

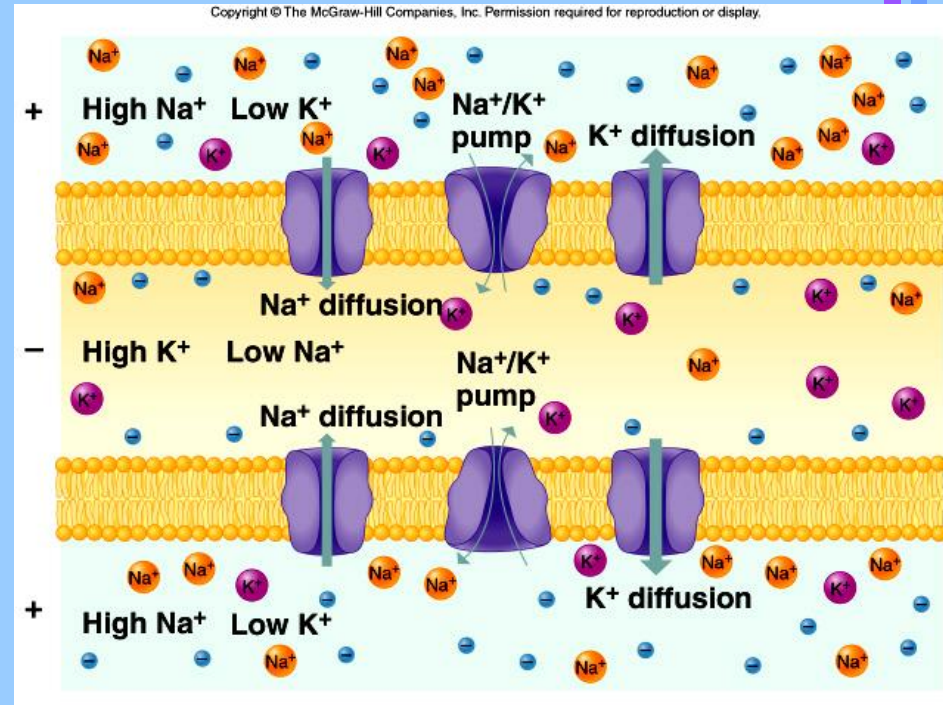
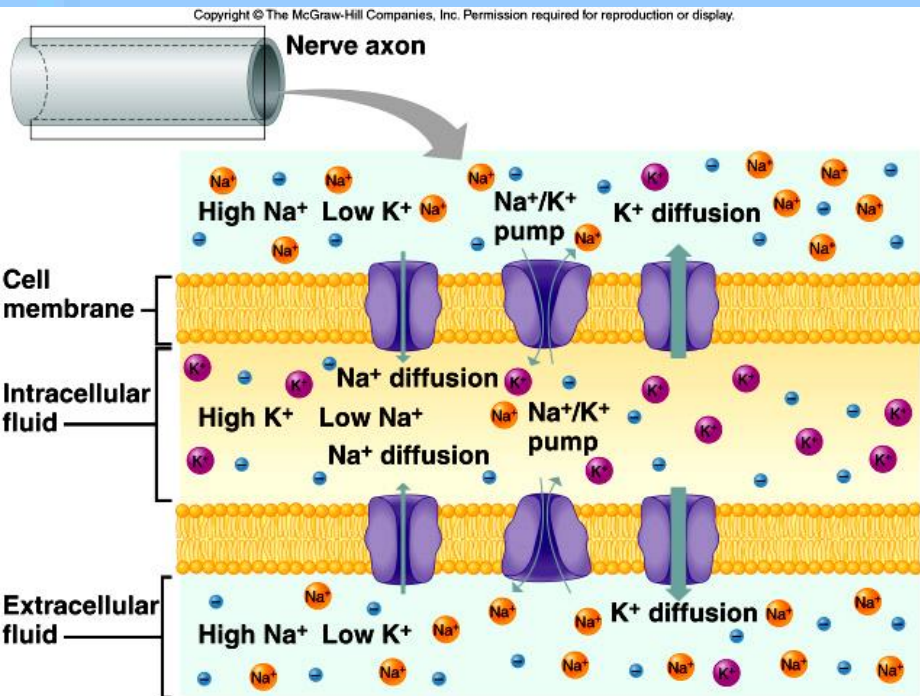
Erhlander and Gasser's classification

- Nerve fibers are classed into three types – **group A nerve fibers, group B nerve fibers, and group C nerve fibers**. Groups A and B are myelinated, and group C are unmyelinated. These groups include both sensory fibers and motor fibers.
- <https://www.youtube.com/watch?v=elbtweWo63M>

- There are four subdivisions of group A nerve fibers: **alpha (α) A α ; beta (β) A β ; , gamma (γ) A γ , and delta (δ) A δ** . These subdivisions have different amounts of myelination and axon thickness and therefore transmit signals at different speeds.

✴ **Cell Membrane Potential**

- A. A cell membrane is usually polarized, with an excess of negative charges on the inside of the membrane; polarization is important to the conduction of nerve impulses.



B. Distribution of Ions

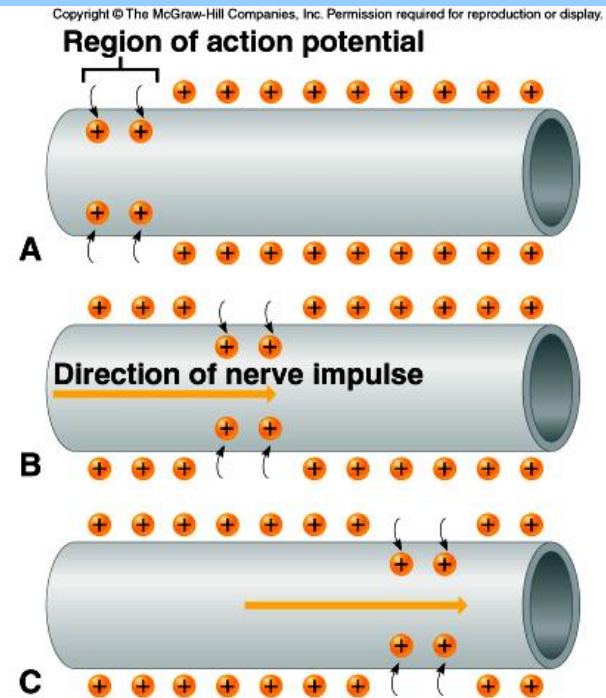
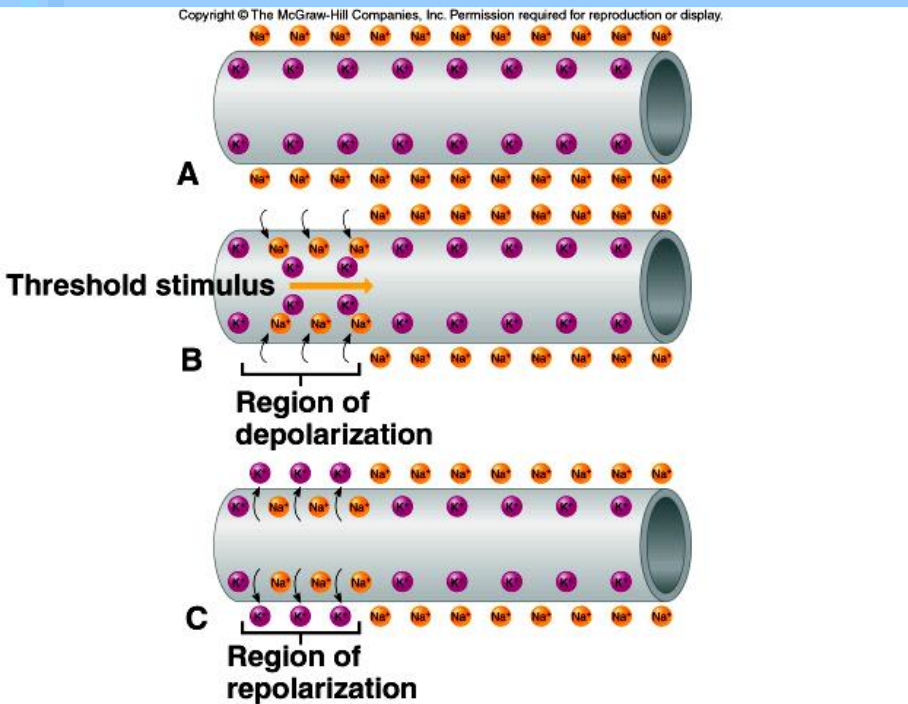
1. The distribution of ions is determined by the membrane channel proteins that are selective for certain ions.
2. Potassium ions pass through the membrane more readily than do sodium ions, making potassium ions a major contributor to membrane polarization.

C. **Resting Potential**

1. Due to active transport, the cell maintains a greater concentration of sodium ions outside and a greater concentration of potassium ions inside the membrane.
2. The inside of the membrane has excess negative charges, while the outside has more positive charges.
3. This separation of charge, or potential difference, is called the resting potential.

D. **Potential Changes**

1. Stimulation of a membrane can locally affect its resting potential.
2. When the membrane potential becomes less negative, the membrane is depolarized.
3. If sufficiently strong depolarization occurs, a threshold potential is achieved as ion channels open.



4. At threshold, action potential is reached.
5. Action potential may be reached when a series of subthreshold stimuli summate and reach threshold.

E. Action Potential

1. At threshold potential, membrane permeability to sodium suddenly changes in the region of stimulation.
2. As sodium channels open, sodium ions rush in, and the membrane potential changes and becomes depolarized.

3. At the same time, potassium channels open to allow potassium ions to leave the cell, the membrane becomes repolarized, and resting potential is reestablished.
4. This rapid sequence of events is the action potential.
5. The active transport mechanism then works to maintain the original concentrations of sodium and potassium ions.

✧ Nerve Impulse

_____ A. A nerve impulse is conducted as action potential is reached at the trigger zone and spreads by a local current flowing down the fiber, and adjacent areas of the membrane reach action potential.

B. Impulse Conduction

1. Unmyelinated fibers conduct impulses over their entire membrane surface.
2. Myelinated fibers conduct impulses from node of Ranvier to node of Ranvier, a phenomenon called saltatory conduction.
3. Saltatory conduction is many times faster than conduction on unmyelinated neurons.

C. **All-or-None Response**

1. If a nerve fiber responds at all to a stimulus, it responds completely by conducting an impulse (all-or-none response).
2. Greater intensity of stimulation triggers more impulses per second, not stronger impulses.

A social robot is an artificial intelligence (AI) system that is designed to do interaction with humans and other robots. In the workplace, social robots got potential to take over entire job functions, such as greeting and basic customer service. A personal robot can help at home to perform things like cook meals with fresh food, or. Help to cook, by cutting vegetables, mixing things. Manage the fridge and ordering food, with an algorithm like a mind map to reduce waste. Give snacks and drinks to home guests in a party. Cognitive robots get their goals done by perceiving their environment, paying attention to the events that matter, planning what to do, anticipating the outcome of their actions and the actions of other agents, and learning from the resultant interaction through probability concept. They deal with the inherent uncertainty of natural environments by continually learning, reasoning, and sharing their knowledge with global databases. Robot being able to perceive the world from someone else's view point, a cognitive robot can anticipate that person's intended actions and needs. Same applies during direct interaction like, robot assisting a surgeon in hospital theatre.

Are robots in line with our values?



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Dr Kartheek Balapala is A renowned Scholar of Neural physiology. His research interests include: postural changes in blood pressure, impact of brain function on human behaviour. As medical doctor, he published over 14 books in global markets and a firm believer of autodidacticism and metacognition.



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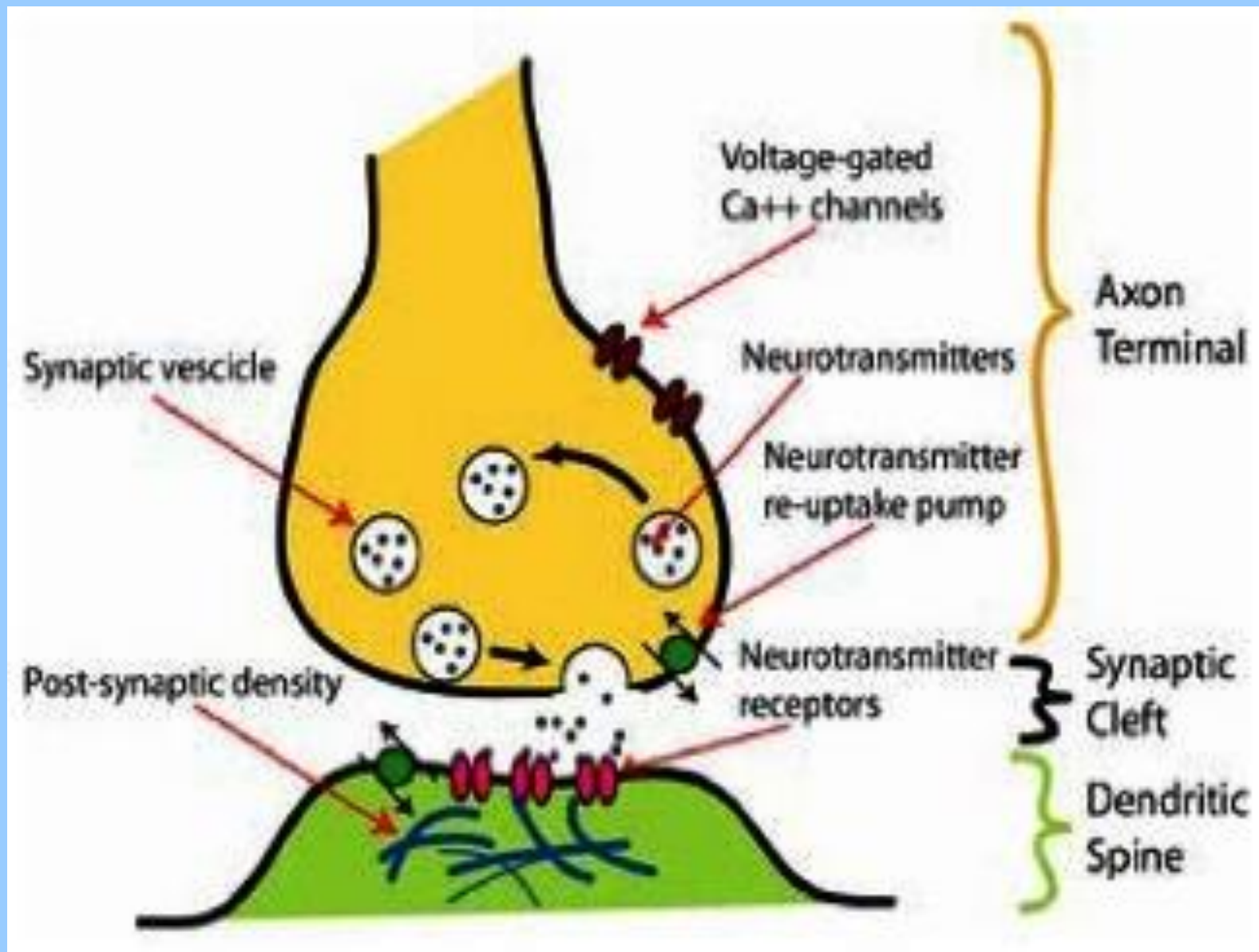
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The Synapse

- A. Nerve impulses travel from neuron to neuron along complex nerve pathways.
- B. The junction between two communicating neurons is called a synapse; there exists a synaptic cleft between them across which the impulse must be conveyed.



All or None law in neurophysiology is basis for propositional logic in Robotics

